
BAYESIAN SMALL AREA ESTIMATION OF UNDER-FIVE MALARIA PREVALENCE IN WEST AFRICA

A PREPRINT

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Abstract

Reliable district-level estimates of malaria prevalence in children under five are essential for targeting intervention resources, monitoring progress against the WHO Global Technical Strategy and evaluating equity, but Demographic and Health Surveys (DHS) are designed to be representative at the regional level rather than at finer administrative scales. We develop a Bayesian hierarchical area-level Fay-Herriot framework with a BYM2 spatial prior to produce smoothed district-level prevalence estimates for under-five malaria across Burkina Faso, Côte d'Ivoire and Ghana, using two DHS rounds per country spanning the 2010 to 2022 period. The modelling sequence builds up from a single-country cross-sectional baseline to country-level space-time models, then to four three-country specifications that pool spatial structure across national borders on a combined 724-district adjacency graph. A formal sensitivity analysis assesses the cross-border adjacency assumption, the time-invariance of covariate slopes, and the assumption of shared versus country-specific BYM2 hyperparameters using leave-one-out cross-validation. The cross-sectional baseline empirically prefers the BYM2 prior over both pure ICAR and IID alternatives. The country-level and cross-border models recover three distinct trajectories: substantial declines in Burkina Faso and Ghana, and a marked increase in Côte d'Ivoire between 2012 and 2021. None of three structural relaxations of the headline cross-border specification produces a detectable improvement in expected log predictive density, so the simpler shared-BYM2 calendar-year specification is parsimoniously adequate and is retained as the headline model.

Keywords small area estimation · Bayesian hierarchical model · BYM2 spatial prior · Fay-Herriot · malaria prevalence · West Africa · Demographic and Health Surveys

1 Introduction

1.1 Background

Malaria remains one of the leading causes of childhood morbidity and mortality in sub-Saharan Africa. The World Health Organization estimated 263 million cases and 597,000 deaths globally in 2023, of which approximately 94 per cent occurred in the African Region [1]. The burden is geographically heterogeneous within countries: districts only short distances apart can differ in prevalence by tens of percentage points,

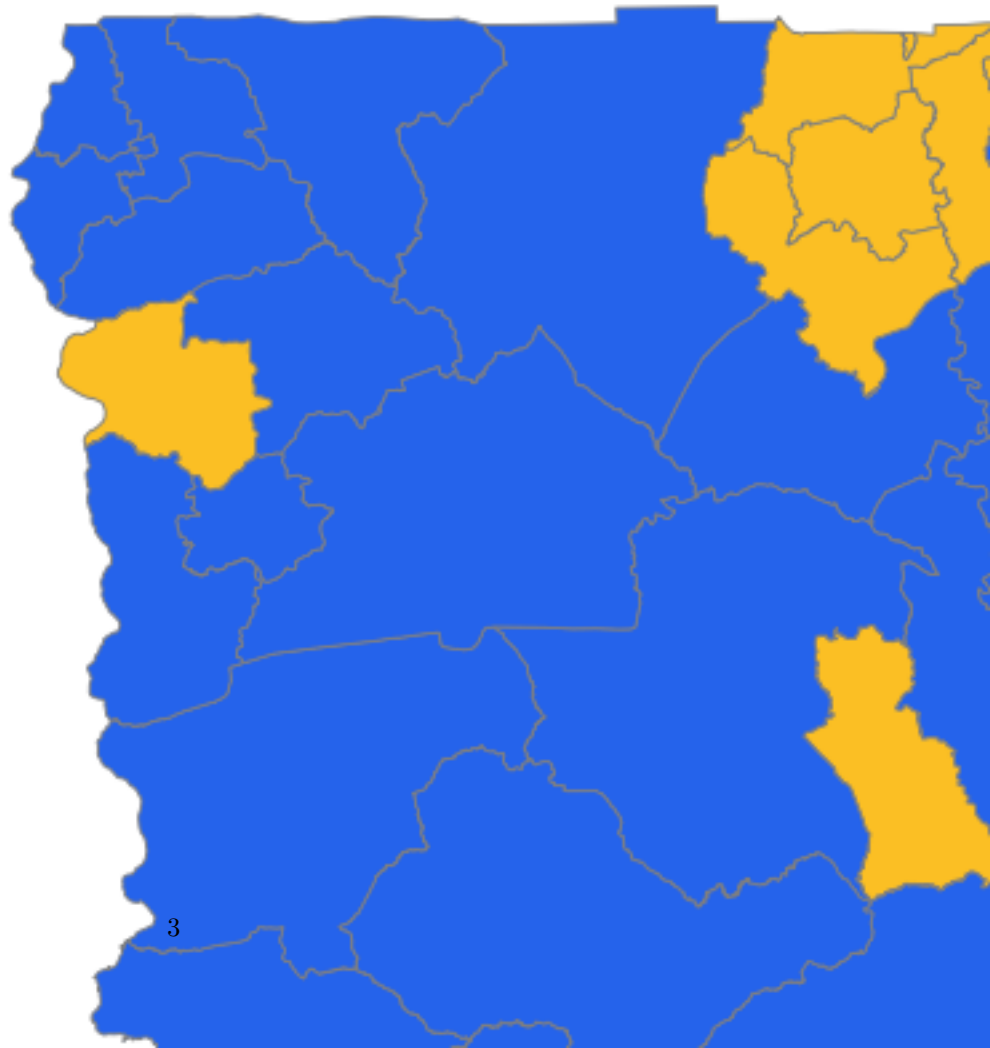
reflecting local variation in vector ecology, climatic suitability, intervention coverage and socioeconomic conditions [2, 3]. Sub-national estimates of prevalence at the district scale are therefore essential for targeting intervention resources, monitoring progress against the WHO Global Technical Strategy for Malaria 2016 to 2030 [1], and evaluating the equity of malaria control programmes.

Demographic and Health Surveys (DHS) are one of the most widely used sources of nationally representative malaria prevalence data in sub-Saharan Africa, with parasite rapid diagnostic tests (RDTs) conducted on children aged 6 to 59 months in standardised survey designs [4]. The DHS sampling design uses stratification by administrative region and stratified two-stage cluster sampling, with districts (admin-2 units) generally not used as explicit strata. Direct survey-based estimates at the district level are therefore based on a small number of clusters per district and have sampling variances too large for reliable interpretation [5, 6]. In our data, many districts are sampled with too few clusters to permit a stable design-based estimate, and a substantial number of districts are not sampled at all in a given round. Figure 1 illustrates this for the three study countries and the two survey rounds, and Figure 2 provides a zoomed view of one country alongside its administrative regions to make the granularity of the district sampling pattern apparent. The patchwork of sampled and unsampled districts is the analytical problem this work addresses.

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1.2 Statistical problem and motivation

Small area estimation (SAE) addresses the problem of producing reliable estimates for areas in which the direct sample is too small to support a stable design-based estimator. The literature distinguishes between unit-level SAE, which models the individual response, and area-level SAE, which models a transformed summary statistic from each area. The Fay-Herriot model [7] is the canonical area-level approach. It treats the design-based direct estimate from each area as a noisy observation of an underlying true value with sampling variance treated as known [8], and links the unobserved true value to area covariates and area-level random effects.

For prevalence outcomes, modelling on the logit scale ensures the latent quantity respects the unit interval and gives a linear hierarchical model [9]. Spatial dependence between areas is typically captured by an intrinsic conditional autoregressive (ICAR) prior or its convolution variants [10], with the BYM2 reparameterisation of [11] providing identifiability between the structured and unstructured components and consistent interpretation of the spatial standard deviation. The Bayesian implementation in Stan [12, 13] is now standard for this class of model.

This work builds and compares a sequence of Bayesian area-level SAE models for under-five malaria prevalence across three West African countries. Each country has been surveyed twice during the 2010 to 2022 period, giving a two-time-point space-time data structure across 724 districts. The principal methodological contribution is a careful investigation of cross-border spatial pooling: whether and how the three country panels should be joined on a single adjacency graph that includes cross-border edges.

1.3 Multi-country pooling and the cross-border question

A defining choice in this work is whether to pool the three countries on a single combined adjacency graph that includes cross-border edges. The substantive case for pooling is that malaria transmission ecology does not respect national borders. *Plasmodium falciparum* dominates transmission in West Africa and shares *Anopheles gambiae* and *Anopheles funestus* vector species across the three study countries [1]. The West African monsoon drives a regional July to October transmission peak and a north to south climatic gradient that the political borders cut transversely [2]. Mobile human populations disperse the parasite across borders, and the structural determinants of risk (climate, elevation, urbanisation, accessibility) are themselves spatially smooth across the region [3].

At the same time, joining three countries on a single graph raises a methodological question about the temporal alignment of the surveys. The first survey rounds span four calendar years (Burkina Faso 2010, Côte d’Ivoire 2012, Ghana 2014), and a spatial smoother that treats cross-border edges in the first round as if they connected contemporaneous surfaces would conflate spatial with temporal variation [14]. The cross-border modelling sequence addresses this concern through a layered strategy. A three-country single-time model uses only the temporally aligned second survey round, removing the temporal misalignment by construction. A three-country two-time model extends to both rounds with country-specific linear slopes on calendar year, following [6]. Two further specifications, plus a direct cross-border adjacency sensitivity check, test the structural assumptions of the two-time model. The relative merits of each specification are assessed by leave-one-out cross-validation [15] and consolidated as a formal sensitivity analysis.

1.4 Outline

Section 2 sets out the data sources and statistical methods. Section 3 presents the results of the seven fitted specifications and the sensitivity comparison. Section 4 discusses the findings, their relation to the literature, the implications for practice, and the limitations of the work. Section 5 concludes.

2 Methods

2.1 Data

2.1.1 Survey instruments and outcome

The data are drawn from the DHS programme [4]. The outcome is the result of an RDT for *Plasmodium falciparum* infection administered to children aged 6 to 59 months. Six DHS surveys are used in this study (Table 1). The administrative geography is the second-level administrative division (admin-2), referred to as district in Ghana and Côte d’Ivoire and as commune (Round 2) or province (Round 1) in Burkina Faso.

The harmonised country-time district panel contains $N = 724$ districts (Ghana 260, Burkina Faso 351, Côte d’Ivoire 113). A small number of districts changed boundaries between rounds; these were handled by majority-area cross-walks documented in the methodological workbook.

Table 1: DHS surveys used in this study, by country and round.

Country	Survey round	Fieldwork window	Districts sampled
Burkina Faso	DHS 2010	May 2010 to Jan 2010	45
Burkina Faso	MIS 2021	Nov 2021 to Mar 2022	70
Côte d’Ivoire	DHS 2011-12	Dec 2011 to May 2012	33
Côte d’Ivoire	DHS 2021	Sep 2021 to Dec 2021	33
Ghana	DHS 2014	Sep 2014 to Dec 2014	207
Ghana	DHS 2022	Oct 2022 to Jan 2023	256

2.1.2 Direct estimation and sampling variance

For each district i in survey round t , the design-based direct estimate of malaria prevalence is the Hájek estimator [16],

$$\hat{p}_{it} = \frac{\sum_{k \in S_{it}} w_k m_k}{\sum_{k \in S_{it}} w_k}, \quad (1)$$

where S_{it} is the set of sampled children in district i at round t , w_k is the survey weight of child k , and $m_k \in \{0, 1\}$ is the RDT result. The Hájek estimator is preferred over the Horvitz-Thompson estimator because the denominator is also estimated from the survey, which reduces variability when sample sizes within domains are small [17, 18]. The design-based sampling variance is computed by Taylor-series linearisation [19] using a `svydesign` specification with primary sampling unit and stratum information from the DHS recode files. The logit transform $y_{it} = \text{logit}(\hat{p}_{it})$ gives a quantity on the unbounded real line; the associated sampling variance on the logit scale follows from the delta method,

$$\psi_{it}^2 = \frac{\widehat{\text{Var}}(\hat{p}_{it})}{[\hat{p}_{it}(1 - \hat{p}_{it})]^2}. \quad (2)$$

Direct estimates of exactly zero or one are handled by a small continuity adjustment; districts with zero observed clusters in a round are flagged as unobserved and enter the model only through the spatial and covariate structure.

2.1.3 Covariates

Seven district-level covariates are included in the cross-border models. The covariate set is selected to capture the dominant ecological and structural determinants of malaria transmission identified in the regional literature [2, 3]. Covariate values for each district-round were extracted using the actual DHS fieldwork months of that survey, so that seasonal variation in rainfall and temperature is captured at the data level rather than through an explicit seasonality term in the model. Each covariate was standardised to unit variance and zero mean prior to model fitting.

The covariates were assembled from three distinct data sources, each operating at a different native spatial resolution. The first source is remote-sensed raster surfaces: elevation from the SRTM 1km digital elevation model, rainfall from CHIRPS daily precipitation [3], land surface temperature from the MODIS LST product, and nighttime lights from the VIIRS annual composite [20]. Each raster was clipped to district polygons and aggregated to the district mean (or median in the case of travel time) using the official district boundaries. The travel-time-to-the-nearest-city covariate is taken from the global friction surface of [3]. Population density is computed from the WorldPop population counts divided by district land area, log-transformed to handle the heavy right tail.

The second source is country-specific census tabulations for the poverty covariate. Ghana publishes a district-level multidimensional poverty index through the Ghana Statistical Service StatsBank API at the admin-2 level, which was queried directly for each Ghanaian district. Burkina Faso publishes its poverty data at the regional (admin-1) level through the *Recensement Général de la Population et de l’Habitation* (RGPH 2019), and Côte d’Ivoire publishes its multidimensional poverty index at the regional (admin-1) level through the *Enquête sur le Niveau de Vie des Ménages* (ENV 2015). For these two countries, the regional poverty value was assigned uniformly to every district within the region.

The third source is the official admin-2 shapefile from each country for boundary information and derived covariates. Table 2 lists each covariate, its substantive mechanism for the malaria-prevalence relationship and its data source. The full data-preparation pipeline is documented in the methodological workbook.

Table 2: Covariates used in the cross-border models.

Covariate	Mechanism	Source	Reference
Elevation	Altitude lowers ambient temperature and slows parasite development inside <i>Anopheles</i> vectors.	SRTM 1km DEM	\citep{bhatt2015}
Rainfall	Rainfall sustains vector breeding sites; wet-season totals are a standard environmental driver.	CHIRPS daily precip.	\citep{weiss2020}
Temperature	Temperature governs vector survival and the extrinsic incubation period; optimum near 25 C.	MODIS LST	\citep{bhatt2015}
Nighttime lights	Urbanisation reduces larval habitat and improves housing and healthcare access.	VIIRS composite	\citep{elvidge2014}
Travel time	Accessibility from major centres reflects the practical reach of interventions and treatment.	Friction surface	\citep{weiss2020}
Pop. density	Density modifies effective transmission intensity and human-vector contact rates.	WorldPop + shapefile	\citep{who2023}
Poverty	Household-level deprivation drives ITN ownership, care-seeking and housing quality.	Census, district (GHA), region (BFA, CIV)	\citep{floreay2014}

Caveat on covariate resolution. A material caveat applies to the poverty covariate. Ghana publishes district-level poverty figures, but Burkina Faso and Côte d’Ivoire only publish equivalent measures at the regional (admin-1) level. For these two countries, the regional poverty value is assigned uniformly to every district within the region, so within-region between-district variation in poverty is not captured. The poverty covariate should be interpreted as a region-level determinant for Burkina Faso and Côte d’Ivoire and as a district-level determinant for Ghana. The other six covariates are extracted at the district scale or finer for all three countries. The implication of this caveat is examined in Section 4.5.

2.1.4 Spatial adjacency

A combined country-time-district adjacency graph W was constructed by identifying districts that share a common border. The graph has $N = 724$ nodes and $E = 2,056$ edges in total, of which 1,978 are within-country edges and 78 are cross-border edges. The BYM2 scaling factor [11] was computed on this combined graph for the full cross-border specifications and on the disconnected within-country subgraph (with one component per country) for the within-country-only sensitivity model, following [21].

2.2 Statistical framework

2.2.1 Fay-Herriot likelihood

The Fay-Herriot likelihood [7] treats the logit-transformed direct estimate as a noisy observation of an underlying true latent quantity:

$$y_{it} \sim \mathcal{N}(\theta_{it}, \psi_{it}^2). \quad (3)$$

The sampling variance ψ_{it}^2 is treated as known following [8]. The linking model decomposes the latent quantity as

$$\theta_{it} = \mu + \alpha_{c[i]} + b_i + \eta_{c[i],t} + \gamma_{it} + \mathbf{x}_{it}^\top \boldsymbol{\beta}_{c[i]}, \quad (4)$$

where μ is the global intercept, α_c is a country deviation identified by $\sum_c \alpha_c = 0$, b_i is the district-level spatial random effect, $\eta_{c,t}$ is the temporal component (parameterised differently across models), γ_{it} is an unstructured space-time interaction, and $\boldsymbol{\beta}_c$ is a vector of country-specific covariate slopes. The country-specific covariate slopes are partially pooled across countries through the hierarchical prior $\beta_{c,k} \sim \mathcal{N}(\bar{\beta}_k, \tau_k^2)$ [22].

2.2.2 BYM2 spatial prior

Spatial dependence between districts is modelled by the BYM2 prior [11], a reparameterisation of the convolution model of [10]. The district-level spatial term is

$$b_i = \sigma_b \left(\sqrt{1 - \phi} v_i + \sqrt{\phi/s} u_i \right), \quad (5)$$

with $v_i \sim \mathcal{N}(0, 1)$ the unstructured Gaussian component and $u \sim \text{ICAR}(W)$ the structured intrinsic conditional autoregressive component on the adjacency matrix W . The ICAR density is

$$p(u \mid W) \propto \exp \left(-\frac{1}{2} \sum_{(i,j) \in E} (u_i - u_j)^2 \right), \quad (6)$$

where E is the set of edges of W . The sum-to-zero constraint $\sum_i u_i = 0$ identifies u . The scaling factor s is computed from the generalised inverse of the structure matrix following [11] so that σ_b has consistent marginal-SD interpretation regardless of the adjacency structure. For graphs with multiple disconnected components the scaling factor is computed per component [21]. The mixing parameter $\phi \in (0, 1)$ governs the relative contribution of structured and unstructured spatial variation: $\phi = 0$ recovers a pure IID specification and $\phi = 1$ recovers a pure ICAR specification; treating ϕ as an unknown parameter lets the data determine the mixture [13].

2.2.3 Priors

Weakly informative priors are used throughout, with the priors on the spatial standard deviation following the penalised-complexity (PC) framework of [23]. The shared prior specification across the cross-border models is given in Table 3.

Table 3: Prior distributions for the shared parameters across the cross-border models.

Parameter	Prior	Rationale	Source
μ	$\mathcal{N}(-1.85, 1.5^2)$	Centred on logit of moderate paediatric prevalence	\cite{gelman2006}
α_c (free)	$\mathcal{N}(0, 1.5^2)$	Identifies free elements under sum-to-zero	\cite{gelman2006}
σ_b	$\text{Exp}(4.6)$	PC, $P(\sigma_b > 1) \approx 0.01$	\cite{simpson2017}
ϕ	$\text{Beta}(3, 2)$	Mild preference for structured component	\cite{morris2019}
σ_γ	$\text{Exp}(2)$	PC on space-time interaction SD	\cite{simpson2017}
$\bar{\beta}_k$	$\mathcal{N}(0, 1)$	Weakly informative on standardised covariates	\cite{gelman2006}
τ_k	$\text{Exp}(2)$	PC on between-country slope SD	\cite{simpson2017}
$\bar{\delta}$	$\mathcal{N}(0, 0.5^2)$	Tight on regional annual rate of change	\cite{wakefield2020}
σ_δ	$\text{Exp}(5)$	PC on between-country slope variation	\cite{simpson2017}

2.2.4 Computation and diagnostics

All models were fitted in Stan version 2.34 [12] via the `cmdstanr` interface, using the No-U-Turn Sampler [24]. Four chains were run in parallel with 1,500 warmup and 2,000 sampling iterations each. Cross-sectional and country-level models used `adapt_delta = 0.99` and `max_treedepth = 14`; cross-border models used `adapt_delta = 0.97` and `max_treedepth = 13`. Convergence was assessed by the rank-normalised $\hat{R}$ of [25], bulk and tail effective sample sizes, the number of divergent transitions reported by the sampler, and visual inspection of trace, rank and pairs plots. Non-centred parameterisations were used for all hierarchical components to mitigate funnel geometry [26]. Model comparison used approximate leave-one-out cross-validation [15] via the Pareto-smoothed importance sampling implementation in the `loo` package.

2.3 Model sequence

This section sets out the full mathematical specification of each model fitted. The notation is shared throughout: i indexes districts, $t \in \{1, 2\}$ indexes survey rounds, $c \in \{1, 2, 3\}$ indexes countries ($c = 1$ Ghana, $c = 2$ Côte d’Ivoire, $c = 3$ Burkina Faso), and k indexes covariates.

2.3.1 Model 1: Cross-sectional Ghana baseline

Model 1 fits the Fay-Herriot likelihood to Ghana 2022 alone, in three variants distinguished by the spatial prior. The common likelihood and linking model is

$$y_i \sim \mathcal{N}(\theta_i, \psi_i^2), \quad \theta_i = \mu + b_i + \mathbf{x}_i^\top \boldsymbol{\beta}, \quad (7)$$

with $\mathbf{x}_i$ the standardised seven-covariate vector for district i . The variants differ in the form of b_i : IID with $b_i \sim \mathcal{N}(0, \sigma_b^2)$, pure ICAR with $b_i \sim \text{ICAR}(W_{\text{GHA}})$, and BYM2 with b_i as in equation 5. The three variants are compared by LOO-CV [15] to motivate the spatial prior choice for the subsequent models.

2.3.2 Model 2: Country-level space-time models

Model 2 fits a country-specific Fay-Herriot space-time model independently to each country. For country c ,

$$\theta_{it} = \mu_c + b_i + \delta_{c,t} + \gamma_{it} + \mathbf{x}_{it}^\top \boldsymbol{\beta}_c, \quad (8)$$

where $\delta_{c,1} = 0$ identifies μ_c as the Round 1 mean and $\delta_{c,2}$ is the round-to-round shift on the logit scale, $\gamma_{it} \sim \mathcal{N}(0, \sigma_\gamma^2)$ is an unstructured space-time interaction following the Type I classification of [27], and b_i is BYM2 with country-specific scaling factor on the within-country adjacency graph.

2.3.3 Model 3: Three-country single-time

Model 3 is the first cross-border model. It joins all three countries on the combined 724-district adjacency graph W but uses only the temporally aligned second survey round (Burkina Faso 2021, Côte d'Ivoire 2021, Ghana 2022). With every district surveyed within a one-year window, cross-border smoothing operates on contemporaneous surfaces by construction:

$$\theta_i = \mu + \alpha_{c[i]} + b_i + \mathbf{x}_i^\top \boldsymbol{\beta}_{c[i]}, \quad (9)$$

with b_i BYM2 on W , $\sum_c \alpha_c = 0$, and $\boldsymbol{\beta}_{c,k} \sim \mathcal{N}(\bar{\beta}_k, \tau_k^2)$.

2.3.4 Model 4: Three-country two-time, calendar year

Model 4 extends Model 3 to both rounds. The temporal effect is parameterised as a linear slope on the actual calendar year of each survey, centred at 2016:

$$\theta_{it} = \mu + \alpha_{c[i]} + b_i + \delta_{c[i]} \cdot \tilde{\text{yr}}_{it} + \gamma_{it} + \mathbf{x}_{it}^\top \boldsymbol{\beta}_{c[i]}, \quad (10)$$

with $\tilde{\text{yr}}_{it} = \text{yr}_{it} - 2016$ and $\delta_c \sim \mathcal{N}(\bar{\delta}, \sigma_\delta^2)$. This parameterisation follows the discrete spatio-temporal framework of [6] in which temporal effects represent calendar-time rates of change rather than wave-to-wave jumps; the between-round odds ratio for country c is $\exp(\delta_c \cdot \Delta \text{yr}_c)$ where Δyr_c is the country's actual elapsed years. The single time-invariant BYM2 spatial field assumes that the residual spatial structure within each country is stable across the 2010 to 2022 window. This assumption is empirically tested in Model 4a, in the edge-deletion sensitivity, and in Model 4b.

2.3.5 Model 4a: Time-varying covariate slopes

Model 4a relaxes the time-invariance of the covariate slopes in Model 4:

$$\theta_{it} = \mu + \alpha_{c[i]} + b_i + \delta_{c[i]} \cdot \tilde{\text{yr}}_{it} + \gamma_{it} + \mathbf{x}_{it}^\top \boldsymbol{\beta}_{c[i],t[i]}, \quad (11)$$

with $\boldsymbol{\beta}_{c,t,k} \sim \mathcal{N}(\bar{\beta}_k, \tau_k^2)$, giving 42 slope parameters partially pooled around seven regional means. The substantive question is whether the covariate-prevalence relationships are stable across the study window at district scale, or whether they have shifted between rounds.

2.3.6 Model 4b: Country-specific BYM2 components

Model 4b replaces the shared BYM2 hyperparameters of Model 4 with country-specific ones, following the disconnected-graph framework of [21] and the country-specific BYM2 implementation of [28]. The cross-border edges are removed from the adjacency graph, and each country gets its own variance $\sigma_{b,c}$ and mixing parameter ϕ_c :

$$b_i = \sigma_{b,c[i]} \left(\sqrt{1 - \phi_{c[i]}} v_i + \sqrt{\phi_{c[i]}/s_{c[i]}} u_i \right), \quad (12)$$

with u ICAR on the within-country adjacency W' , per-country sum-to-zero $\sum_{i \in c} u_i = 0$, and per-country scaling factor s_c . The substantive question is whether the three countries have materially different residual spatial structures.

2.3.7 Edge-deletion sensitivity

A direct sensitivity analysis to the cross-border edges themselves follows [29]: Model 4 is refitted with the cross-border edges removed but with the BYM2 hyperparameters held constant. This isolates the contribution of the 78 cross-border edges, holding the rest of the specification fixed. The scaling factor is computed per component [21] and applied per district.

2.4 Model comparison

Models within comparable observation blocks are compared by approximate leave-one-out cross-validation [15]. The expected log predictive density (ELPD) is the criterion; larger ELPD indicates better expected predictive performance. Differences in ELPD between models are compared against the standard error of the difference; a difference smaller than two standard errors is considered not statistically distinguishable from zero [15]. Comparisons are only made between models fitted to the same set of observations.

3 Results

3.1 Cross-sectional baseline: motivating the BYM2 prior

The three Ghana 2022 cross-sectional variants converged with all parameters showing $\hat{R} < 1.02$ and bulk effective sample sizes above 300. The LOO-CV comparison is summarised in Figure 3. The BYM2 specification achieves the best ELPD with an estimate of -104.3 (SE 7.9). The IID specification is 2.7 ELPD units behind (SE 1.8), and the pure ICAR specification is 8.9 ELPD units behind (SE 4.2). The ICAR penalty exceeds two standard errors, so the data provide clear evidence against the pure ICAR prior at this scale; the IID alternative is within 1.5 standard errors of BYM2, but the BYM2 parameter estimates indicate that the data prefer a substantial structured component. The BYM2 prior is adopted as the spatial prior for all subsequent models.

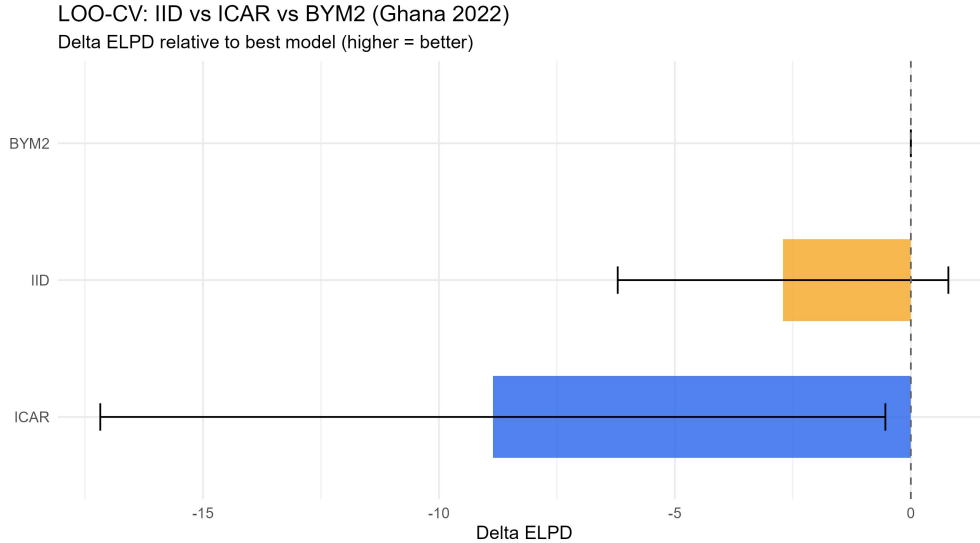


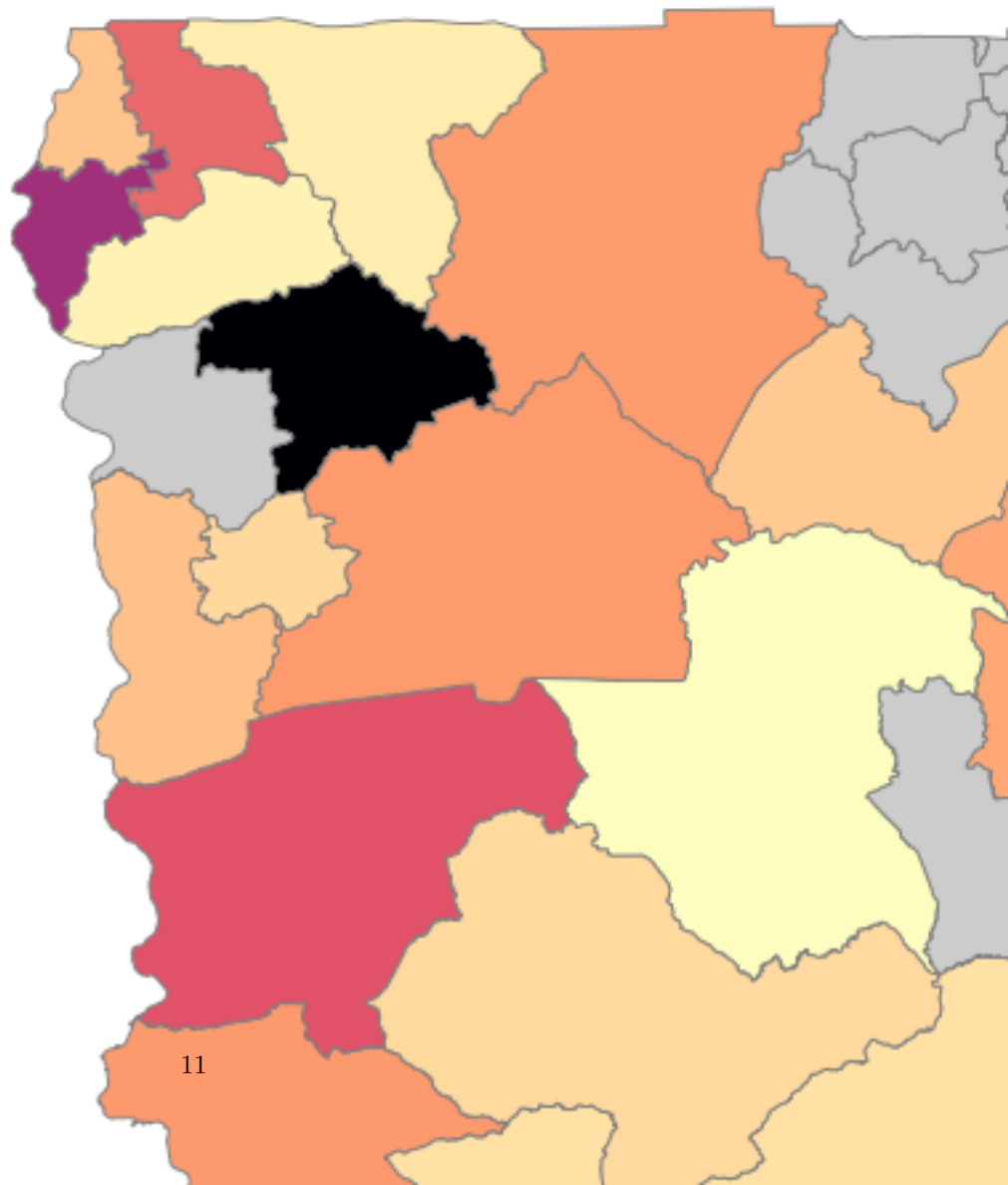
Figure 3: Leave-one-out cross-validation comparison of three spatial priors on the Ghana 2022 cross-section. Bars show the ELPD difference relative to the best model with two-standard-error intervals. The BYM2 prior is preferred over the IID alternative and substantially over the pure ICAR alternative.

The Model 1 BYM2 posterior estimates of the structural parameters are $\mu = -1.82$ (90 per cent CrI $[-2.02, -1.62]$), $\sigma_b = 0.88$ $[0.68, 1.15]$, and $\phi = 0.44$ $[0.17, 0.72]$. The marginal national prevalence is $p^{\text{nat}} = 14.1$ per cent $[11.8, 16.5]$, consistent with the direct national estimate for Ghana 2022. The maps of direct, IID, ICAR and BYM2 estimates for Ghana 2022 are shown in Figure 4. The BYM2 surface reproduces the broad north-to-south gradient visible in the direct estimates while filling in the unsampled districts with spatially coherent predictions.

Model comparison:

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3.2 Country-level space-time models

The country-level Model 2 fits converged for all three countries with $\hat{R} < 1.01$ for all top-level parameters and bulk effective sample sizes between 600 and 5,500. The key parameter estimates are summarised in Table 4 and the implied national prevalence trajectories are shown in Table 5.

Table 4: Model 2 (country-level space-time) posterior estimates of structural parameters. Posterior median with 90 per cent CrI.

Country	μ	σ_b	ϕ	σ_{γ}	δ_2
GHA	-0.83 [-0.94, -0.71]	0.72 [0.56, 0.94]	0.47 [0.20, 0.73]	0.52 [0.45, 0.59]	-1.14 [-1.30, -0.98]
BFA	0.91 [0.74, 1.07]	1.04 [0.78, 1.36]	0.64 [0.37, 0.84]	0.51 [0.45, 0.58]	-2.52 [-2.82, -2.19]
CIV	-1.56 [-1.75, -1.38]	0.82 [0.55, 1.23]	0.59 [0.25, 0.87]	0.48 [0.40, 0.56]	0.78 [0.50, 1.06]

Table 5: Implied national paediatric prevalence by round, country-level Model 2. Posterior medians with 90 per cent CrI.

Country	Round 1	Round 2	Odds ratio R2/R1
GHA	30.5% [28.1, 33.1]	12.3% [10.9, 13.8]	0.32
BFA	71.2% [67.7, 74.5]	16.8% [14.3, 19.5]	0.08
CIV	17.4% [14.8, 20.1]	31.3% [27.9, 34.8]	2.13

The country-level models show three different trajectories. Ghana declined from 30.5 per cent in 2014 to 12.3 per cent in 2022, an odds ratio of 0.32. Burkina Faso showed the largest decline in absolute terms, from 71.2 per cent in 2010 to 16.8 per cent in 2021, with the largest country-level σ_b reflecting substantial residual heterogeneity across districts. Côte d'Ivoire showed an increase from 17.4 to 31.3 per cent between 2012 and 2021. Figure 5 shows the posterior covariate effect estimates from the three country-level models.

Cova

Poster

Elevation

Rainfall

3.3 Three-country cross-border models

The four cross-border specifications were fitted to the full district-round panel. Their key structural parameters are summarised in Table 6.

Table 6: Cross-border model family: key structural parameter estimates. Posterior medians with 90 per cent CrI. Model 4b has per-country spatial parameters: $\sigma_b = (0.15, 0.11, 0.66)$ and $\phi = (0.62, 0.63, 0.72)$ for GHA, CIV, BFA respectively.

Parameter	Model 4	Model 4a	Edge-del	Model 4b
μ	-1.14 [-1.33, -0.95]	-1.15 [-1.34, -0.95]	-0.68 [-1.32, 0.27]	-1.02 [-1.27, -0.42]
σ_b	0.66 [0.40, 0.91]	0.70 [0.45, 0.91]	0.49 [0.25, 0.72]	see note
ϕ	0.85 [0.57, 0.96]	0.86 [0.63, 0.97]	0.79 [0.46, 0.95]	see note
σ_{γ}	0.71 [0.66, 0.76]	0.70 [0.64, 0.75]	0.73 [0.67, 0.78]	0.72 [0.67, 0.77]
δ_{GHA}	-0.143 [-0.166, -0.120]	-0.137 [-0.163, -0.118]	-0.142 [-0.165, -0.118]	-0.142 [-0.166, -0.120]
δ_{CIV}	0.080 [0.049, 0.110]	0.076 [0.047, 0.106]	0.079 [0.049, 0.110]	0.080 [0.050, 0.108]
δ_{BFA}	-0.230 [-0.255, -0.206]	-0.232 [-0.254, -0.205]	-0.233 [-0.256, -0.210]	-0.229 [-0.254, -0.205]
$\bar{\beta}_{\text{NTL}}$	-0.19 [-0.43, 0.04]	-0.17 [-0.27, 0.10]	-0.19 [-0.43, 0.10]	-0.21 [-0.46, 0.07]

3.3.1 Headline findings from Model 4

The Model 4 estimates are consistent with Model 2 in sign and approximate magnitude. The global intercept is $\mu = -1.14$ on the logit scale. The spatial standard deviation $\sigma_b = 0.66$ is well identified, and the mixing parameter $\phi = 0.85$ indicates that the residual spatial variation is predominantly structured. The country-year slopes recover the three trajectories at calendar-year scale: Ghana $\delta = -0.143 [-0.166, -0.120]$ per year; Côte d’Ivoire $\delta = +0.080 [0.049, 0.110]$ per year; Burkina Faso $\delta = -0.230 [-0.255, -0.206]$ per year. The implied between-round odds ratios are 0.32 for Ghana (8-year window), 2.09 for Côte d’Ivoire (9-year window) and 0.08 for Burkina Faso (11-year window), all consistent with Model 2. The strongest regional protective effect is for nighttime lights with $\bar{\beta} = -0.19$, followed by population density at -0.18 and temperature at -0.14 . Elevation, rainfall and travel time have regional credible intervals that span zero, and poverty has a regional median of $+0.09$.

3.3.2 Model 4a: time-varying covariate slopes

Model 4a’s structural parameters are nearly identical to Model 4’s, and the country-year slopes recover the same trajectories. The Round 1 versus Round 2 covariate slopes are stable for almost all of the 21 country-covariate pairs. The largest changes between rounds are seen in the Ghana temperature slope (R1 -0.22 , R2 -0.11) and Ghana population density slope (R1 -0.22 , R2 -0.14), but in both cases the Round 2 credible intervals widen and the change is within the noise. Figure 6 displays the time-varying slopes.

Covariate

Posterior m

Travel time

Temperature

Rainfall

3.3.3 Edge-deletion sensitivity

Removing the 78 cross-border edges and re-fitting Model 4 yields the edge-deletion specification. The country-year slopes are essentially unchanged ($\delta_{\text{GHA}} = -0.142$, $\delta_{\text{CIV}} = 0.079$, $\delta_{\text{BFA}} = -0.233$), as are the regional covariate effects. The spatial standard deviation drops from $\sigma_b = 0.66$ to $\sigma_b = 0.49$, reflecting that some of the spatial variation previously absorbed by the cross-border smoothing is redistributed into the unstructured Gaussian term and the country intercepts, with the global intercept becoming weakly identified at $\mu = -0.68$. The mixing parameter $\phi = 0.79$ remains high.

3.3.4 Model 4b: country-specific BYM2 components

Model 4b yields country-specific spatial standard deviations of $\sigma_{b,\text{GHA}} = 0.15$, $\sigma_{b,\text{CIV}} = 0.14$ and $\sigma_{b,\text{BFA}} = 0.66$. Burkina Faso shows substantially larger residual spatial variation than the other two countries, consistent with its larger country-level Model 2 σ_b . The country-specific mixing parameters are $\phi_{\text{GHA}} = 0.61$, $\phi_{\text{CIV}} = 0.63$ and $\phi_{\text{BFA}} = 0.72$, all indicating predominantly structured residual variation. The country-year slopes are again essentially unchanged from Model 4.

3.4 Sensitivity comparison: LOO across the cross-border models

Table 7 presents the leave-one-out cross-validation comparison across the four cross-border two-time specifications, all fitted on the same observation set so the comparison is valid in the sense of [15].

Table 7: Leave-one-out cross-validation across the cross-border two-time models. ELPD difference is computed against the best model; differences smaller than two standard errors are not statistically distinguishable from zero [15].

Model	ELPD diff	SE diff	ELPD	SE(ELPD)
Model 4a (time-varying betas)	0.0	0.0	-465.2	21.2
Edge-deletion (within-country only)	-2.3	4.9	-467.5	21.3
Model 4 (shared BYM2, full graph)	-3.4	4.9	-468.6	21.1
Model 4b (country-specific BYM2)	-5.9	5.3	-471.2	21.5

Model 4a achieves the best ELPD by a small margin. Model 4 is 3.4 ELPD units behind Model 4a with a standard error of 4.9 units (0.7 standard errors). The edge-deletion specification is 2.3 ELPD units behind (0.5 standard errors), and Model 4b is 5.9 ELPD units behind with a standard error of 5.3 (1.1 standard errors). All four specifications are within 6 ELPD units of each other against standard errors of around 5, and none of the three structural relaxations of Model 4 produces a detectable improvement in expected predictive performance.

4 Discussion

4.1 Headline findings

The Bayesian space-time Fay-Herriot framework with BYM2 spatial prior produced district-level estimates of under-five malaria prevalence for Burkina Faso, Côte d’Ivoire and Ghana over two DHS rounds spanning the 2010 to 2022 period. The cross-sectional baseline confirmed that the BYM2 prior is preferred over the IID and ICAR alternatives by LOO-CV, supporting its use across the subsequent models. The country-level and three-country specifications agree on three distinct trajectories. Ghana and Burkina Faso showed substantial declines, with Burkina Faso’s decline the largest in absolute terms (from 71 to 17 per cent), while Côte d’Ivoire showed an increase from 17 to 31 per cent between 2012 and 2021. The headline cross-border specification (Model 4) recovers these trajectories on a calendar-year scale with country-specific annual rates of change.

The mixing parameter ϕ from Model 4 is 0.85 with credible interval well above 0.5, indicating that the residual spatial variation after conditioning on the seven structural covariates is predominantly structured. This is consistent with the substantive expectation that the structural covariates capture the broad regional gradient and the spatial residual reflects unmeasured but spatially smooth factors such as intervention history, housing quality and vector species composition.

4.2 Sensitivity analysis and the cross-border assumption

Three structural relaxations of Model 4 were fitted as sensitivity specifications. Model 4a allows covariate slopes to differ between rounds, testing the time-invariance assumption. The Model 4 edge-deletion removes the 78 cross-border edges, testing whether cross-border smoothing is doing useful work. Model 4b replaces the shared BYM2 hyperparameters with country-specific ones, testing the assumption that the three countries share spatial structure.

The LOO-CV comparison in Table 7 shows that none of the three relaxations produces a detectable improvement over Model 4. The posterior parameter comparisons in Table 6 reinforce this: the country-year slopes are essentially identical across all four specifications; the regional covariate effects are similar within Monte Carlo error; and the Round 1 versus Round 2 covariate slopes in Model 4a do not differ materially for the great majority of country-covariate pairs. Substantively, Model 4 is parsimoniously adequate against all three structural alternatives, and the simpler specification is retained.

The edge-deletion result deserves attention. Removing the cross-border edges substantially weakens identification of the global intercept μ and produces a corresponding widening of the country intercepts α_c , but the substantive parameters of interest (country-year slopes, regional covariate effects, between-country slope variation) are essentially unchanged. The cross-border edges in Model 4 are doing identification work for the global intercept and the country intercepts, but the data-driven findings about trajectories, covariate effects and between-country heterogeneity do not depend on them. This is a useful negative result: it tells us that the substantive findings are robust to whether the spatial smoother shares information across national borders.

4.3 Relation to the literature

The country-specific trajectories are broadly consistent with the published literature. The Burkina Faso decline matches the trajectory documented in [2] and updated in [3] for the Sahelian zone, where the rollout of seasonal malaria chemoprevention from 2014 onwards has been associated with substantial reductions in paediatric prevalence. The Ghana decline matches the trajectory in the country’s own national malaria control programme reports through the WHO World Malaria Report series [1]. The Côte d’Ivoire increase between 2012 and 2021 is less commonly emphasised in summary regional reports but is consistent with the country’s own DHS reports and with the broader observation in [3] that progress against malaria has not been uniform across West Africa.

The methodological framework is in the tradition of [9, 6] for area-level SAE of paediatric prevalence outcomes, with the BYM2 prior of [11] and the disconnected-graph treatment of [21]. The multi-country pooling on a combined adjacency graph is, to the best of our knowledge, one of the first applications of this design in this region.

4.4 Implications for practice

The district-level estimates produced by the model are suited to target resource allocation for malaria intervention programmes within each country and across the region. The substantially smaller uncertainty of the model-based estimates compared with the direct estimates, particularly for the many districts not directly sampled in a given round, makes them useful as inputs to programme planning. The between-round trajectory information from Model 4 quantifies the rate of progress at the country level on a comparable calendar scale across the three countries, allowing direct comparison of progress that the round-indicator parameterisation would obscure.

4.5 Limitations

Several limitations should be borne in mind.

First, the poverty covariate is available at the district scale only for Ghana. For Burkina Faso and Côte d’Ivoire, regional (admin-1) poverty values were assigned uniformly to every district within the region, so within-region variation in this covariate is not captured. This is a known limitation of nationally available poverty data and reflects the resolution at which the underlying surveys are designed to be representative. The implication is that the poverty covariate coefficient should be interpreted as a region-level determinant for Burkina Faso and Côte d’Ivoire, and that any residual within-region variation correlated with poverty is absorbed into the spatial random effect.

Second, the BYM2 spatial field is assumed to be time-invariant in Models 3, 4, 4a and 4b. The substantive justification is that the structural drivers of the regional malaria gradient (climate, elevation, urbanisation, accessibility) move on decadal timescales [2, 3], so the spatial residual after conditioning on the covariates should also be time-stable at the district scale across the study window. Fitting a separate BYM2 field per round with only two time points would be weakly identified and was not attempted. With more survey rounds this assumption could be relaxed through the Knorr-Held Type II or Type IV interactions [27, 30].

Third, the sampling variance ψ_{it}^2 is treated as known following the standard Fay-Herriot approach [8]. For districts with very few clusters the design-based variance estimate is itself uncertain; the Fay-Herriot framework does not propagate this uncertainty. Smoothing the variance estimate through a generalised variance function is a natural extension.

Fourth, the cross-border pooling assumes that the cross-border edges represent shared transmission ecology across borders. While there is a strong substantive case for this from vector ecology and human movement [2, 3, 1], the data in this particular application do not strongly select the cross-border specification over the within-country-only alternative. This does not undermine the broader methodological case for cross-border pooling but suggests that applications with longer time series or more countries would be informative.

Fifth, external validation against the Malaria Atlas Project [2, 3] surfaces was set out in the methodological workbook but the comparison itself was not completed within this report. The MAP comparison would provide concordance evidence with an independently developed geostatistical modelling tradition and is the natural next step.

5 Conclusions

This work has presented a Bayesian area-level small area estimation framework for under-five malaria prevalence across Burkina Faso, Côte d’Ivoire and Ghana over the 2010 to 2022 period. The BYM2 spatial prior was selected empirically over the IID and ICAR alternatives in a cross-sectional baseline and adopted for the country-level and cross-border models. The country-level results showed three different trajectories: substantial declines in Burkina Faso and Ghana, and an increase in Côte d’Ivoire, with these findings carrying over to the cross-border specifications. The headline cross-border model (Model 4) parameterises country-specific trajectories as annual rates of change on calendar year, supporting direct between-country comparison.

A formal sensitivity analysis tested three structural relaxations: time-varying covariate slopes (Model 4a), removal of cross-border edges (edge-deletion), and country-specific BYM2 hyperparameters (Model 4b). None produced a detectable improvement in LOO-CV, and the substantive parameters of interest were essentially unchanged across all four specifications. Model 4 is parsimoniously adequate against these alternatives and is retained as the headline specification.

The findings are consistent with the published regional literature on malaria trajectories in West Africa, and the methodological framework provides a transparent basis for producing district-level paediatric prevalence estimates with quantified uncertainty for the three study countries. The framework is directly applicable to other Sahelian and Guinean countries where the DHS RDT modules have been conducted, and extends to other outcomes available in the DHS recode files.

References

- [1] World Health Organization. World malaria report 2023. Technical report, World Health Organization, Geneva, 2023.
- [2] Samir Bhatt, D. J. Weiss, E. Cameron, D. Bisanzio, B. Mappin, U. Dalrymple, et al. The effect of malaria control on plasmodium falciparum in africa between 2000 and 2015. *Nature*, 526(7572):207–211, 2015.
- [3] D. J. Weiss, Tim C. D. Lucas, Michele Nguyen, Anita K. Nandi, Donal Bisanzio, Katherine E. Battle, et al. Mapping the global prevalence, incidence, and mortality of plasmodium falciparum, 2000–17: a spatial and temporal modelling study. *The Lancet*, 394(10195):322–331, 2019.
- [4] Trevor N. Croft, Aileen M. J. Marshall, Courtney K. Allen, et al. Guide to dhs statistics. Technical report, ICF, Rockville, Maryland, USA, 2018. Demographic and Health Surveys Program.
- [5] J. N. K. Rao and Isabel Molina. *Small Area Estimation*. John Wiley & Sons, Hoboken, NJ, 2nd edition, 2015.

- [6] Jon Wakefield, Taylor Okonek, and Jon Pedersen. Small area estimation for disease prevalence mapping. *International Statistical Review*, 88(2):398–418, 2020.
- [7] Robert E. Fay and Roger A. Herriot. Estimates of income for small places: An application of James-Stein procedures to census data. *Journal of the American Statistical Association*, 74(366a):269–277, 1979. doi: 10.1080/01621459.1979.10482505.
- [8] Yong You and J. N. K. Rao. A hierarchical Bayes estimation of survey unemployment rates with general linear mixed models. *Survey Methodology*, 28(1):19–29, 2002.
- [9] Laina D. Mercer, Jon Wakefield, Athena Pantazis, Angelina M. Lutambi, Honorati Masanja, and Samuel Clark. Space-time smoothing of complex survey data: small area estimation for child mortality. *The Annals of Applied Statistics*, 9(4):1889–1905, 2015.
- [10] Julian Besag, Jeremy York, and Annie Mollié. Bayesian image restoration, with two applications in spatial statistics. *Annals of the Institute of Statistical Mathematics*, 43(1):1–20, 1991.
- [11] Andrea Riebler, Sigrunn H. Sørbye, Daniel Simpson, and Håvard Rue. An intuitive bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4):1145–1165, 2016.
- [12] Stan Development Team. *Stan Modeling Language Users Guide and Reference Manual*, 2024. URL <https://mc-stan.org>. Version 2.35.
- [13] Mitzi Morris, Katherine Wheeler-Martin, Dan Simpson, Stephen J. Mooney, Andrew Gelman, and Charles DiMaggio. Bayesian hierarchical spatial models: Implementing the Besag York Mollié model in Stan. *Spatial and Spatio-temporal Epidemiology*, 31:100301, 2019. doi: 10.1016/j.sste.2019.100301.
- [14] Enrique M. Saldarriaga. Hiv-prevalence mapping using small area estimation in kenya, tanzania, and mozambique at the first sub-national level. *arXiv preprint arXiv:2010.03114*, 2020. URL <https://arxiv.org/abs/2010.03114>.
- [15] Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and Computing*, 27(5):1413–1432, 2017.
- [16] Jaroslav Hájek. Comment on “an essay on the logical foundations of survey sampling, part one”. *Foundations of Statistical Inference*, page 236, 1971.
- [17] William G. Cochran. *Sampling techniques*. Wiley, New York, 3rd edition, 1977.
- [18] Thomas Lumley. *Complex Surveys: A Guide to Analysis Using R*. John Wiley & Sons, Hoboken, NJ, 2010.
- [19] David A. Binder. On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review*, 51(3):279–292, 1983.
- [20] Christopher D. Elvidge, Kimberly E. Baugh, Mikhail Zhizhin, and Feng-Chi Hsu. Why viirs data are superior to dmsp for mapping nighttime lights. *Proceedings of the Asia-Pacific Advanced Network*, 35:62–69, 2013.
- [21] Anna Freni-Sterrantino, Massimo Ventrucci, and Håvard Rue. A note on intrinsic conditional autoregressive models for disconnected graphs. *Spatial and Spatio-temporal Epidemiology*, 26:25–34, 2018.
- [22] Andrew Gelman. Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, 1(3):515–534, 2006. doi: 10.1214/06-BA117A.
- [23] Daniel Simpson, Håvard Rue, Andrea Riebler, Thiago G. Martins, and Sigrunn H. Sørbye. Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science*, 32(1):1–28, 2017.
- [24] Matthew D. Hoffman and Andrew Gelman. The no-U-turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593–1623, 2014.
- [25] Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718, 2021. doi: 10.1214/20-BA1221.
- [26] Michael Betancourt and Mark Girolami. Hamiltonian Monte Carlo for hierarchical models. *Current trends in Bayesian methodology with applications*, 79(30):2–4, 2015.
- [27] Leonhard Knorr-Held. Bayesian modelling of inseparable space-time variation in disease risk. *Statistics in Medicine*, 19(17-18):2555–2567, 2000.

- [28] Connor Donegan. Flexible functions for ICAR, BYM, and BYM2 models in Stan, 2021. URL <https://github.com/ConnorDonegan/Stan-IAR>. Code repository, GitHub.
- [29] Arul Earnest, Geoffrey Morgan, Kerrie Mengersen, Louise Ryan, Richard Summerhayes, and John Beard. Evaluating the effect of neighbourhood weight matrices on smoothing properties of conditional autoregressive (CAR) models. *International Journal of Health Geographics*, 6:54, 2007. doi: 10.1186/1476-072X-6-54.
- [30] Tomas Goicoa, Aritz Adin, María Dolores Ugarte, and James S. Hodges. In spatio-temporal disease mapping models, identifiability constraints affect pql and inla results. *Stochastic Environmental Research and Risk Assessment*, 32(3):749–770, 2018.